Hybrid Graph Neural Network and Collaborative Filtering Based Warehouse Storage Optimization for Efficient Order Picking

by

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Certificate by Supervisor

This is to certify that the project report entitled "Hybrid Graph Neural Network and Collaborative Filtering Based Warehouse Storage Optimization for Efficient Order Picking" submitted by Mr. Prakash Kumar Shah (Roll No.: B220022CS), a student of the Department of Computer Science and Engineering, National Institute of Technology Sikkim, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech), is a record of original work carried out by him under my supervision and guidance.

The work presented in this report has not been submitted, either in part or full, to any other University or Institute for the award of any degree or diploma.

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Certificate by Student

I hereby declare that the project report entitled "Hybrid Graph Neural Network and Collaborative Filtering Based Warehouse Storage Optimization for Efficient Order Picking" is a bonafide record of the work carried out by me under the supervision of Dr. Krishna Kumar, Department of Computer Science and Engineering, National Institute of Technology Sikkim.

I further declare that this work has not been submitted, either in part or full, for the award of any other degree or diploma to any University or Institute.

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Dated: 02-06-2026	**Mr. Prakash Kumar Shah**
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Date: 02-06-2026

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List of Acronyms

ABC	Activity-Based Classification
AHC	Agglomerative Hierarchical Clustering
CF	Collaborative Filtering
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GNN	Graph Neural Network
GraphSAGE	Graph Sample and Aggregate
K-Means	K-Means Clustering
ReLU	Rectified Linear Unit
SKU	Stock Keeping Unit
S-Shape	S-Shape Routing Strategy
t-SNE	t-distributed Stochastic Neighbor Embedding
UK	United Kingdom

Abstract

This project presents a hybrid graph-based and collaborative filtering approach for warehouse storage optimization aimed at minimizing picker travel distance during order fulfillment. The proposed framework utilizes historical transaction data from the Online Retail dataset to extract SKU-level demand features such as purchase frequency, customer count, sales contribution, and demand intensity. A co-purchase graph is constructed to model relationships between products that are frequently purchased together.

A Graph Convolutional Network (GCN) is employed to learn meaningful product embeddings from the co-purchase graph and SKU features. In addition, collaborative filtering techniques are used to capture customer purchasing behavior and product similarity. The hybrid feature representation is obtained by fusing the learned GCN embedding and collaborative filtering features for warehouse storage optimization.

Hybrid feature space is clustered using K-Means clustering algorithm to group similar products into storage zones and store frequently co-picked products closer. To evaluate the effectiveness of the proposed approach, simulated warehouse layouts are used with multiple order-picking strategies and routing methods such as S-Shape and Midpoint routing.

The experimental results indicate that the proposed hybrid approach is superior to the traditional clustering-based approaches such as Agglomerative Hierarchical Clustering (AHC) in terms of average picker travel distance. The results presented here show that combining graph-based relational learning with customer purchasing behavior provides a significant benefit to the efficiency of storage assignment and overall warehouse performance.

Overall, we demonstrate the effectiveness of Graph Neural Networks and collaborative filtering for intelligent warehouse storage optimization and efficient order picking.

Chapter 1

Introduction

1.1 Background

Warehouses play a crucial role in contemporary supply chains and logistics. Their main purpose is the storing of products and the processing and delivery of customers' orders. Given the fast development of online shopping websites and e-commerce in general, the need for warehouses to work with a wider range of products and a higher number of customer orders is growing. The complexity of operations in such facilities makes their efficient and intelligent storage necessary.

The process of order picking, which implies the collection of certain products located in various storage areas to satisfy the customers' needs, is one of the central operations in the warehouses. In this process, proper product organization inside the facility plays a decisive role. That is why storage assignment is very important for warehouse performance.

In the context of modern retail and warehouses, transaction data are generated in large quantities, and they contain valuable information about customer purchase behavior. Specifically, transaction data may show how often products are purchased jointly and what is the relationship between products. This information may be used to analyze the behavior of customers and optimize warehousing. It is still hard to extract information from transactional datasets and make storage decisions on the basis of this information.

The recent advancements in machine learning and graph learning give a chance to solve the mentioned problem in an innovative way. Graph neural networks are capable of capturing the relations between products, because the latter can be represented as a set of nodes in a graph. Additionally, collaborative filtering approaches can detect the hidden relationships between products, because they can recognize the similarity in purchase behavior of customers. Thus, this kind of technology allows one to make data-driven decisions on storing products in warehouses.

The main idea of the project is to use transaction data analysis, graph learning, collaborative filtering, and clustering to optimize warehousing of products.

1.2 Motivation

This project is motivated by the rising trend in improving warehouse efficiency using effective product storage assignment. In most warehouses, products are assigned according to demand rate, frequency of sales, or certain other pre-determined product storing rules. Such strategies are useful, but they ignore the rich information that is embedded within the data generated by customer purchases.

In natural scenarios, customer purchase history creates relationships between various products. Some products have a high tendency to be purchased alongside another product, whereas some products tend to have common purchasing behaviors amongst different customers. All these factors generate valuable insights into product organization within the warehouse space, which can be exploited to design efficient product organization layouts.

Additionally, there is currently the availability of state-of-the-art machine learning algorithms that can learn complicated relationships among products. Classical storage assignment strategies fall short of modeling these relationships effectively. Using graph neural networks, it is possible to learn representations for different products using co-purchasing networks. On the other hand, collaborative filtering can be used to model customer purchasing preferences and behavioral similarities.

The motivation behind this project is the use of the combination of transactional data analysis, graph learning, and customer analysis to develop a methodology for efficient storage organization in warehouses. The methodology would help in finding the connections among the different types of products, putting similar items together in zones that are most suited to the items and allocating the zones into the appropriate areas within the warehouse.

1.3 Challenges

Although many studies have been carried out on warehouse storage assignment and order-picking optimization, several important challenges still exist.

- **Limited Consideration of Product Relationships**

Most existing storage assignment methods mainly focus on product demand, sales frequency, or inventory turnover. However, they often do not consider whether products are frequently purchased together, which is important for creating an efficient storage layout.

- **Difficulty in Understanding Product Associations**

  Customer transaction data contains valuable information about relationships between products. Extracting these relationships and understanding how products are connected through purchasing patterns remains a challenging task.

- **Separate Analysis of Products and Customers**

  Many existing approaches focus either on product relationships or customer purchasing behavior. Combining both types of information to obtain a better understanding of product associations is still not widely explored.

- **Limited Use of Graph-Based Learning in Warehouses**

  Graph Neural Networks have been successfully applied in recommendation systems and supply chain applications. However, their use in warehouse storage assignment and storage zone generation is still limited.

- **Creating Storage Zones Based on Real Purchasing Patterns**

  Traditional storage assignment approaches often rely on fixed rules or static clustering methods. Developing storage zones that reflect actual customer purchasing behavior and product relationships remains an important challenge.

- **Need for More Intelligent Storage Assignment Methods**

  Existing methods do not fully utilize the information available in transaction data. There is a need for intelligent approaches that can learn product relationships automatically and use them to create more effective warehouse storage layouts.

1.4 Problem Statement

Warehouse storage assignment is a critical task in warehouse management because it determines how products are organized and accessed during order fulfillment. The arrangement of products within the warehouse has a direct impact on the efficiency of order-picking operations, which represent a major portion of warehouse activities. An effective storage layout should ensure that products are placed in locations that support smooth and efficient retrieval while making the best use of available storage space.

Modern warehouses store a large number of products and handle thousands of customer transactions. These transactions contain valuable information about customer purchasing behavior, particularly the relationships between products that are frequently purchased together. Such relationships provide important insights into how products should be organized within the warehouse. However, identifying and utilizing these relationships for storage assignment is a complex problem due to the large volume of transaction data and the interconnected nature of product interactions.

The conventional storage assignment approach is largely based on the attributes like product demand, frequency of sales, turnover ratio, or predetermined warehouse storage rules. Although this kind of strategy works in terms of managing warehouse operations, it fails to take into account the inherent relations between the products in the transaction records. This means that the products that have similar purchasing patterns end up being placed inappropriately within the warehouse space due to lack of sufficient consideration of customers' purchase trends.

Furthermore, an aspect that cannot be overlooked in this case is product relationships. Products can be associated directly and indirectly with each other within the network formed by their purchase patterns. Conventional approaches fail to address this problem as well because it is hard to represent the product relationship. In this regard, what needs to be developed is a framework that is able to learn product representation and product interactions within the purchasing network.

Finally, once relationships have been established, there is a need for storage assignment strategies for the products. It is necessary to group related products together and place them in appropriate storage locations within the warehouse space.

Hence, the problem being addressed by this research project is the design of a smart warehouse storage allocation scheme which can analyze the relationship between different products based on past transaction history, represent these relationships efficiently, and make use of these relationships in designing optimal storage areas within warehouses. This framework will be able to arrange products based on relationships and purchasing history.

1.5 Objectives

The primary objective of this research is to build a smart warehouse storage optimization system that uses historical transaction data to organize products more efficiently. The goal is to learn about customer purchase patterns, relationships between different products, and use this knowledge to optimize their storage within the warehouse. By placing products that are often purchased together close to each other, the proposed system would make it easier to pick orders faster and more efficiently.

The objectives of this research can be summarized as follows:

- Analyze historical transaction data to learn about purchasing behavior. The purpose of this task is to find out which products are often purchased together and how the relationship between products looks like.
- Build a network of product relationships based on patterns observed in transaction data. Such a product graph provides an organized representation of product relationships based on actual customer transactions.
- Use Graph Neural Networks to derive meaningful representations of products from their graph structure. Product representations are learned to incorporate direct and indirect relationships between products that cannot be discovered using traditional approaches.
- In order to integrate Collaborative Filtering methods in order to model similarities in customers' buying behavior and increase our understanding of the connections between different products.
- In order to combine the knowledge gained through Graph Neural Networks and Collaborative Filtering into a single representation of the relationships between the products.
- In order to cluster products that have similar attributes and patterns of purchase into storage zones optimized for their particular properties. This would result in logical product clusters being formed inside the warehouse.
- In order to decide which storage locations are suitable for a certain type of product depending on the relationships and clusters obtained previously.
- In order to develop the most optimal warehouse layout taking into account the actual patterns of purchasing behavior rather than using only traditional factors like order frequency and volume.
- In order to optimize order picking process through creation of an efficient storage scheme.
- In order to show how data driven decision making can be used in real world problems of warehouse optimization and storage assignment.

1.6 Chapter Summary

This chapter introduced the problem of warehouse storage optimization and highlighted the importance of reducing picker travel distance in modern warehouse systems. With the rapid growth of e-commerce and digital supply chain systems, warehouses are required to operate more efficiently while minimizing operational cost. Among different warehouse activities, order picking is one of the most time-consuming and labor-intensive operations, where a significant portion of the operational cost is caused by the travel distance covered by workers during product collection.

The chapter discussed how storage location assignment plays an important role in improving warehouse efficiency and reducing unnecessary picker movement. Traditional storage assignment methods such as ABC classification, demand-based storage, and clustering approaches mainly organize products according to demand frequency and sales volume. However, these methods generally do not consider the relationships between the products that are purchased together frequently, which results in inefficient routes for picking and increased operational costs.

This chapter proposed the Hybrid Graph Neural Network and Collaborative Filtering based warehouse storage optimization framework to address these limitations. We propose a system which uses transaction data to build a co-purchase graph and then uses a Graph Convolutional Network (GCN) to learn meaningful relationships between products. Collaborative Filtering (CF) also captures customer purchasing similarity and behavioral patterns. These learned features are combined to generate optimized storage zones using clustering techniques.

The chapter further elaborates on the major objectives of the project such as transaction data preprocessing, feature extraction, graph construction, GCN-based learning, hybrid feature generation, clustering-based storage assignment and warehouse performance evaluation with different picking and routing strategies.

This chapter set the stage for the proposed warehouse optimization framework and emphasized the significance of graph-based learning techniques for smart storage assignment and minimizing the travel distance of pickers.

Chapter 2

Literature Review

This chapter presents the review of existing literature on storage location assignment, order-picking optimization, and data-based solutions for efficient warehouse operation. In light of the increasing popularity of online businesses, the minimization of picker travel distances and storage assignment have gained prominence in the field of research.

The conventional strategies for storage allocation take into account factors like product demands, inventory turnover rate, clustering, and product associations. For example, Zhou et al. [1] suggested the utilization of clustering and association rule mining in assigning storage locations. Products were categorized according to several attributes, and associated products were stored close together. This method enhanced picking performance and minimized travel distances. However, the strategy was highly dependent on the transaction patterns from historical data.

The more recent studies include the works by Kalbha et al. [2] and Zarinchang et al. [3], who developed storage assignment models based on time-series clustering and order-picking/worker-safety optimization. These methods increased the overall efficiency of warehouses by grouping products in terms of their demand. Nonetheless, none of these methods attempted to capture the inter-product relationship through machine learning algorithms.

The application of machine learning algorithms to warehouse optimization was considered by other researchers as well. For example, Boysen et al. [4] suggested the use of deep reinforcement learning for the dynamic assignment of storage locations in warehouses. Similarly, Liu et al. [5] employed the use of supervised learning together with tree search for real-time storage allocations in robotic fulfillment warehouses. While these approaches offer intelligent solutions to warehousing issues, they may require a significant amount of training time.

In addition, studies have looked into order-picking optimizations within warehouses. Li et al. [6] introduced a framework for joint order batching and picker routing to minimize traveling distance. In a similar fashion, Özbaltan et al. [7] developed a hybrid symbolic control and metaheuristic learning framework for cluster order picking in robotic warehouses. Their approach involved optimization of robot traveling distance using intelligent routing techniques. However, none of these works considered issues related to storage location assignment and product relationships.

Graph Neural Networks (GNNs) are new and promising methodologies that learn from graph-structured data. The effectiveness of GNNs has been discussed in a survey paper by Wu et al. [8]. Using the advantages of GNNs, Tu et al. [9] used GNNs for recommending suppliers in massive supply chains, and Cvancara [10] utilized graph-based learning algorithms for optimizing the locations of facilities. However, despite these advances, the application of GNNs in warehouse storage location assignment has received little attention yet.

The existing methodologies mainly include demand analysis, clustering algorithms, association rules mining, routing optimization, and machine-learning-based decision-making. While these methods contribute towards improving the performances of warehouses, they fail to consider the complexities of relationships between products within transactions. This paper aims to fill this gap by using Graph Neural Network algorithms for modeling product co-purchases and determining storage locations for maximizing warehouse efficiency and reducing picker travel distances.

2.1 Related Tools and Technologies

Data-driven and intelligent methods have become popular in improving warehouse performance and minimizing the time for order picking. Storage location allocation is one of the most crucial processes in a warehouse because the distance traveled by a picker constitutes a significant share of the costs incurred.

The commonly used methods include demand-oriented and clustering-oriented strategies. For instance, Kalbha et al. [2] developed a hybrid storage location assignment strategy based on time-series clustering and found out that it enhanced warehouse performance across different routes. Additionally, Zarinchang et al. [3] created an adaptive storage assignment method that takes into account picking efficiency and worker safety issues. While the two methods improve storage allocation, they do not consider product relationships.

To incorporate the relationships between products, researchers have come up with clustering techniques as well as association rule-oriented approaches. For example, Zhou et al. [1] designed a new storage location allocation scheme based on clustering algorithms and association rule mining. Using clustering analysis, the method identified items frequently purchased together and allocated them storage locations that were close to each other. Thus, picker traveling distances were minimized.

Machine learning algorithms have also been used extensively to optimize warehouse operations. Boysen et al. [4] devised a reinforcement learning model to assign optimal storage locations dynamically, while Liu et al. [5] employed a machine learning and tree search algorithm for optimal storage location assignment in robotic fulfillment centers. However, these approaches rely heavily on extensive data and computation.

Order-picking optimization has also been extensively studied by various researchers in warehouse management. Li et al. [6] proposed a solution that optimized picking and routing of orders. Similarly, Özbaltan et al. [7] developed a symbolic control and bio-inspired metaheuristic based algorithm to optimize the routing process in the context of robotic cluster order picking. Although this work optimized routing and minimized the distance traveled by robots, neither study considered the problem of storage location assignment.

Graph Neural Network (GNN) models have provided fresh insights into capturing complex relationships in graphs. The work by Wu et al. [8] focused on how GNNs can capture both nodes' properties and their relations. On this basis, Tu et al. [9] employed GNNs for the recommendation of suppliers in large supply chains, and Cvancara [10] utilized graph learning for solving problems related to facility location optimization.

Existing approaches use clustering, association rules, machine learning, and optimization models to enhance warehouse efficiency. Most approaches do not take full advantage of the information contained in the relational structure of the products' transactions. Therefore, a Graph Neural Network is suggested to learn the relationship among the products based on their co-purchasing pattern and reduce the distance traveled by pickers.

2.2 Related Works

Various techniques have been suggested by researchers to increase warehouse efficiency by improving storage location assignment and picking process. Traditional solutions concentrate on grouping products depending on their frequency of purchase, sales volume, or other characteristics related to inventory items. Besides, clustering has been extensively applied for grouping products having similar frequency of purchases. For instance, Kalbha et al. [2] suggested a product placement technique using time-series clustering that allowed to enhance the performance of various order-picking methods. Similarly, an adaptive approach for assigning storage locations taking into account efficiency and worker safety was presented by Zarinchang et al. [3]. Despite providing better outcomes in comparison with traditional solutions, the approaches mentioned above concentrate solely on demand characteristics without accounting for relationships between co-purchased products.

To overcome this disadvantage, a number of clustering and association rule based solutions have been developed by researchers. Thus, Zhou et al. [1] suggested a clustering analysis approach along with association rule mining. Placing co-purchased products at close proximity storage locations significantly decreased picker traveling distance and picking time. The algorithm uses pre-defined associations and transaction history for making decisions.

Machine learning has been increasingly applied to warehouse optimization as well. For instance, Boysen et al. [4] proposed a deep reinforcement learning-based framework that facilitates storage location assignment decisions through time-changing warehouse environments. On similar lines, Liu et al. [5] incorporated supervised learning along with tree search algorithms for real-time storage assignments in robotic fulfillment facilities. While such techniques can help in better decision-making, they might require heavy computation power and data training.

In addition to storage decisions, order-picking optimization has also been extensively investigated in the existing literature. In particular, Li et al. [6] proposed a framework to jointly optimize order batching and picker routing problems in robotic warehouses, resulting in less travel distance. Further, recent studies like Özbaltan et al. [7] have integrated the use of symbolic control along with bio-inspired metaheuristic methods for cluster order picking in robotic warehouses. Though this technique optimized robot routes, it did not consider storage location decisions and product relationships.

The interest in Graph Neural Networks (GNN) has increased considerably due to their capability to model complex relationships between connected nodes. According to Wu et al., the GNN can be used to learn not only features of nodes but also relational dependencies [8]. Based on these capabilities, Tu et al. utilized the GNN technique to solve the problem of supplier recommendation for large-scale supply chain systems [9], and Cvancara used a graph approach to facility location optimization [10].

Although significant advances were achieved thanks to current methods, there are several shortcomings that need to be addressed. For example, all warehouse storage assignment techniques depend either on the demand statistics analysis, clustering algorithms, association rules, or optimization models. Thus, the relational information extracted from transaction records is rarely considered. In order to resolve this issue, a novel study is conducted using a Graph Neural Network to learn relationships between products based on the graph built from transactional data and then used with Collaborative Filtering.

Table 2.1: Summary of Literature Review

Sl. No.	Reference	Overview and Approach	Evaluation Results	Limitations
1	Özbaltan et al. (2025)	Proposed a hybrid symbolic control and bio-inspired optimization framework for warehouse robot path planning.	Reduced robot travel distance and improved cluster order-picking efficiency.	Focuses on path planning rather than storage assignment and SKU relationship modeling.
2	Kalbha et al. (2024)	Developed a hybrid storage assignment strategy using time-series clustering techniques for warehouse optimization.	Improved clustering quality and reduced picker travel distance under different routing policies.	Relies on historical demand patterns and does not consider co-purchase relationships.
3	Tu et al. (2024)	Applied Graph Neural Networks to learn supplier relationships in large-scale supply chain networks.	Improved recommendation accuracy by capturing complex supplier interactions.	Not intended for warehouse storage assignment or order-picking optimization.
4	Cvancara (2024)	Developed a GNN-based framework for facility location optimization problems.	Successfully learned spatial and relational information for location decisions.	Focuses on facility-level optimization rather than warehouse storage assignment.
5	Zarinchang et al. (2024)	Proposed an adaptive storage assignment framework considering picking efficiency and worker safety.	Improved storage allocation while balancing operational performance and safety.	Does not explicitly model product relationships or graph-based learning.
6	Boysen et al. (2022)	Used deep reinforcement learning for dynamic storage location assignment.	Produced adaptive storage decisions and reduced warehouse picking effort.	Requires significant training data and computational resources.
7	Liu et al. (2022)	Combined supervised learning and tree-search methods for real-time storage allocation.	Enabled efficient and fast storage assignment decisions in robotic systems.	Primarily designed for robotic fulfillment environments.
8	Wu et al. (2021)	Provided a comprehensive review of Graph Neural Network architectures and applications.	Highlighted the effectiveness of GNNs for learning complex relational patterns.	Survey paper without warehouse-specific implementation.
9	Zhou et al. (2020)	Applied clustering analysis and association rule mining for storage location assignment.	Improved picking efficiency by grouping related and frequently co-purchased products.	Depends on historical transaction data and predefined association rules.
10	Li et al. (2017)	Proposed a joint optimization approach for order batching and picker routing.	Reduced travel distance and improved overall order-picking efficiency.	Focuses on routing optimization rather than storage assignment or product relationships.

2.3 Chapter Summary

This chapter presented an overview of previous researches on storage location assignment problems, order picking optimization and intelligent warehouse management techniques. Commonly, the traditional methodologies, including demand-based storage location assignment, clustering techniques, association rule mining, have been widely utilized to optimize the storage locations of items, which helps to minimize picker movements. For example, Zhou et al. proved that by integrating clustering and association rules, the storage assignment problem can be optimized because commonly bought products would be stored together.

Additionally, this chapter reviewed some recent development in the field of intelligent and adaptive storage location assignment problems. For example, Kalbha et al. introduced a methodology for improving storage assignment decision based on time series clustering, and Zarinchang et al. introduced an adaptive storage assignment approach for improving the efficiency and safety of workers.

Other research has used machine learning and optimization tools to solve warehouse management issues as well. Machine learning tools such as deep reinforcement learning and supervised learning have been utilized to make storage assignments dynamically and instantly, while optimization tools such as routing optimization and order-picking optimization have helped minimize the distance and make operations more efficient. This benefit was proven by two studies conducted by Li et al. and Özbaltan et al.

Finally, the chapter touched upon the increased application of Graph Neural Networks (GNNs) to logistics and supply chain problems. It is evident from existing literature that GNNs are extremely effective in learning interconnections between different entities and helping to make decisions using them. The effectiveness of GNNs was demonstrated in supplier recommendation and facility location optimization. However, little has been done to apply GNNs to warehouse storage location assignment.

On average, the existing methodologies use either demand analytics, clustering, association analysis, optimization of routing process, or machine learning modeling approaches. Though these methodologies help increase performance in warehouses, they fail to leverage all relational information included in product purchase history. It is precisely here that there exists room for leveraging the potential of Graph Neural Networks and building intelligent warehouse storage zones by taking into consideration product relationships based on historical transaction data.

Having analyzed the state-of-the-art research works, in the following chapter, it will be suggested a methodology for optimizing the storage process in a warehouse. Specifically, the suggested method uses Graph Neural Networks together with Collaborative Filtering to analyze product co-purchases and create intelligent storage zones.

Chapter 3

Proposed Works

3.1 Overview of the Proposed Framework

The proposed work presents a Hybrid Graph Neural Network and Collaborative Filtering based warehouse storage optimization framework designed to minimize picker travel distance and improve warehouse order fulfillment efficiency. The framework combines graph-based relational learning, customer purchasing similarity analysis, clustering-based zoning, and warehouse routing simulation to generate intelligent storage assignments for products inside the warehouse.

Unlike traditional storage assignment approaches such as ABC classification and demand-based clustering, the proposed framework does not treat each product independently. Traditional approaches mainly rely on sales frequency or product demand and ignore relationships between products that are frequently purchased together. In practical warehouse environments, customer orders generally contain multiple related products, and storing these products far apart increases picker movement, aisle switching, and operational cost.

To address the limitations of existing methods, the proposed framework models warehouse products and their relationships using a graph-based learning approach. As shown in Figure 3.1, the framework includes transaction preprocessing, SKU feature engineering, co-purchase graph construction, Graph Convolutional Network (GCN) learning, collaborative filtering-based similarity computation, hybrid feature generation, clustering for storage zoning, warehouse position assignment, and simulation-based evaluation of warehouse performance.

3.1 Overview of the Proposed Framework

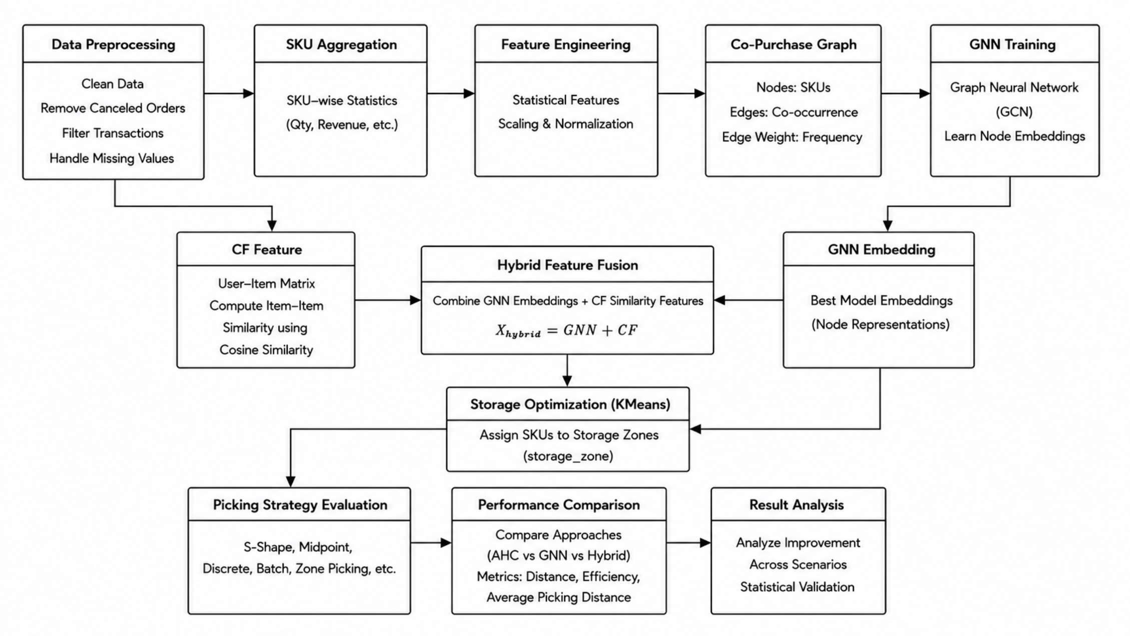

Figure 3.1: Overall Workflow of the Proposed Hybrid GCN and Collaborative Filtering Based Warehouse Storage Optimization Framework

The overall workflow of the proposed warehouse optimization framework is shown in Figure 3.1. First, transaction data is cleaned and transformed into structured information at the SKU level. Then, a co-purchase graph is created to illustrate the relationships between products that are often purchased together. This graph is fed to a Graph Convolutional Network (GCN) to learn product embeddings. In parallel, Collaborative Filtering is also used to learn behavioral similarities based on purchase history. The outputs of both methods are merged to generate hybrid product features, which are used for the assignment of warehouse storage based on clustering.

First, the raw transaction records in the Online Retail dataset are collected and preprocessed. Invalid transactions like cancelled orders, missing customer identifiers, and records with negative or zero quantity values are excluded to improve data quality and consistency. Statistical features are extracted at SKU level from the cleaned dataset after preprocessing. These include total quantity purchased, invoice frequency, number of customers, revenue contribution, lifecycle duration and rate of demand. The extracted features represent the product demand behavior and the operational importance in the warehouse.

Customer orders are then generated using invoice numbers to preserve co-purchase relationships between products. Based on these orders, a co-purchase graph is constructed where each SKU is represented as a node and co-occurrence relationships between products are represented as weighted edges. The edge weight represents how frequently two products appear together in customer transactions.

Table 3.1 presents the statistics of the generated co-purchase graph.

Table 3.1: Co-Purchase Graph Statistics

Graph Property	**Value**
Number of SKU Nodes	788
Number of Graph Edges	263,300
Node Feature Dimension	6
Edge Index Shape	[2, 527388]
Edge Weight Count	527,388

The generated graph structure captures real customer purchasing behavior and co-picking relationships among products. A Graph Convolutional Network (GCN) is applied to the constructed graph to learn meaningful product embeddings. The GCN aggregates neighboring node information using graph convolution operations and generates low-dimensional vector representations that capture both SKU demand characteristics and relational product behavior.

In parallel, Collaborative Filtering (CF) is used to compute customer purchasing similarity between products. A user-item interaction matrix is generated from customer transaction records, and cosine similarity is calculated to identify products preferred by similar customer groups. These similarity features provide additional behavioral information for storage optimization.

The learned GCN embeddings and collaborative filtering similarity features are combined to generate a hybrid feature representation. K-Means clustering is then applied to the hybrid features to create optimized warehouse storage zones. Products belonging to the same cluster are assigned to nearby warehouse positions in order to reduce picker movement and aisle traversal during order fulfillment.

After the storage locations are assigned, warehouse performance is evaluated through a simulation-based picking analysis. Historical customer orders are replayed in a simulated warehouse environment using different picking strategies, including Discrete Picking, Batch Picking, and Zone Picking, along with S-Shape and Midpoint routing methods. The average picker travel distance is used as the main performance measure.

The results show that the proposed Hybrid framework consistently performs better than both Agglomerative Hierarchical Clustering (AHC) and the standalone GNN approach across all routing scenarios. By placing frequently co-picked products closer together, the Hybrid framework is able to reduce picker travel distance more effectively.

Overall, the proposed framework provides a data-driven and graph-based approach to warehouse storage optimization. By combining product demand data, customer purchase behavior and co-purchase relationships, this provides a more intelligent approach to design warehouse storage and improve picking efficiency.

3.2 Dataset Description and Preprocessing

The proposed framework is applied to the real-world e-commerce transaction records in the Online Retail dataset for warehouse storage optimization. The key information available in the data set is invoice number, SKU, product description, quantity purchased, invoice date, unit price, customer ID, and country. These records provide good information about customer buying habits, product demand and product associations that are often sold together.

The dataset is loaded using the Pandas library and undergoes several preprocessing steps to improve the quality of data before analysis. The real-world transaction data usually has errors and inconsistencies, and thus data cleaning is done to ensure reliable feature extraction and graph learning.

Cancelled invoices were excluded because they do not constitute completed purchases. Transactions with missing customer IDs are excluded, as customer-level information is required for collaborative filtering analysis. In addition, records with non-positive quantities or invalid unit prices are removed to ensure accurate demand estimation and warehouse analysis.

The preprocessing operations performed in the project are summarized below:

- Removal of cancelled invoices
- Removal of missing customer IDs
- Removal of negative and zero quantity values
- Removal of invalid unit prices
- Conversion of invoice dates into datetime format
- Standardization of SKU codes

After preprocessing, the cleaned dataset contained 397,884 valid transaction records with 4,273 unique customers and 788 selected high-frequency SKUs used for warehouse optimization.

Table 3.2: Dataset Statistics After Preprocessing

Metric	**Value**
Cleaned Transactions	397,884
Dataset Columns	8
Unique Customers	4,273
Selected SKUs	788
User-Item Matrix Size	4273×788

Sample records from the cleaned dataset are shown in Table 3.3.

Table 3.3: Sample Cleaned Transaction Records

InvoiceNo	StockCode	Description	Qty	Date	Price	CustomerID	Country
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01	2.55	17850	UK
536365	71053	WHITE METAL LANTERN	6	2010-12-01	3.39	17850	UK
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01	2.75	17850	UK
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01	3.39	17850	UK

Customer orders are then generated by grouping products using invoice numbers. Products purchased within the same invoice are treated as co-purchased items, preserving realistic warehouse order structure and customer purchasing behavior. These generated orders are later used for co-purchase graph construction.

The cleaned dataset is further aggregated at the SKU level to extract statistical demand-related features. These extracted features are later used as node features for Graph Convolutional Network learning.

Table 3.4: Extracted SKU-Level Features

Feature	Description
Total Quantity	Total quantity purchased for each SKU
Number of Invoices	Number of invoices containing the SKU
Number of Customers	Unique customers purchasing the SKU
Total Revenue	Revenue generated by the SKU
Lifecycle Duration	Active selling duration of the SKU
Quantity per Day	Average daily demand of the SKU

The total revenue for each SKU is calculated as:

$$Revenue = Quantity \times UnitPrice$$

The lifecycle duration of each SKU is computed using the difference between its first and last transaction dates:

$$LifeDays = LastDate - FirstDate + 1$$

Average daily demand is then calculated as:

$$QtyPerDay = \frac{TotalQuantity}{LifeDays}$$

Then feature extraction, and then the StandardScaler method is used to normalize the SKU feature matrix to improve the stability of Graph Neural Network training and improve the clustering effect. The normalized matrix obtained from the data has six numerical features for each SKU, and it is utilized as the representation of node features in the graph learning process.

From the customer transaction data, a User-Item Interaction Matrix is also constructed, where rows correspond to customers, columns correspond to products, and the values are the quantities purchased. This matrix reflects the behavior of customers in buying. Later, this matrix is used to calculate similarities between products by collaborative filtering based on cosine similarity.

In short, the preprocessing and feature engineering step converts the raw transaction data into structured representations for graph learning, collaborative filtering, clustering-based storage zoning, and warehouse performance evaluation.

3.3 Customer Purchase Relationship Modeling

The modeling of customer purchasing behaviors is a crucial step in the suggested framework for optimizing the warehouse storage process, because through this modeling, the relationship between the products according to their purchasing behaviors by customers can be captured. In the current method, not only product demand is considered but rather how the products are purchased together by the customers themselves. This will help us understand which products are mostly picked together.

The cleaned transaction data are aggregated into customer orders according to invoice numbers, with each invoice representing a single purchase transaction. The products are repeated in the quantities purchased in each order to preserve the realistic demand patterns. This allows the framework to accurately capture the frequency with which products are co-purchased.

The framework models the customer-product relationships in two major approaches:

- Co-purchase relationship modeling
- Collaborative Filtering (CF) similarity modeling

In co-purchase relationship modeling, products that are purchased together in the same customer order are considered related products. The more frequently two products are bought together across multiple orders, the stronger the relationship between them. This information captures real purchase patterns and is then used to construct the product graph for learning using Graph Convolutional Network (GCN).

To measure these relationships, all possible pairs of SKUs are generated from each invoice and their co-occurrence frequencies are counted. The co-purchase relationship between products (i) and (j) is then computed as:

$$w_{ij} = \sum_k I(i, j \in Order_k)$$

where:

- w_{ij} represents the co-purchase frequency between products i and j
- $Order_k$ represents customer order k
- $I(i, j \in Order_k)$ returns 1 if both products appear in the same order

The relationship weights are higher for products that are more often purchased together. These weights are derived from real customer purchase behavior and are useful in identifying products that are likely to be picked together in warehouse operations. The information about such relationship is useful for building the product graph and improving the storage location assignment.

In addition to co-purchase relationships, Collaborative Filtering similarity analysis is performed using customer-product interaction matrices. A User-Item Matrix is constructed where rows represent customers, columns represent products, and matrix values represent purchased quantities.

Table 3.5 presents the dimensions of the generated User-Item Matrix used for Collaborative Filtering similarity computation.

Table 3.5: User-Item Interaction Matrix Statistics

Property	**Value**
Number of Customers	4,273
Number of Selected SKUs	788
Matrix Dimension	4273×788
Interaction Type	Purchased Quantity

Cosine similarity is then applied to measure similarity between products based on customer purchasing behavior:

$$Similarity(i, j) = \frac{R_i \cdot R_j}{|R_i||R_j|}$$

where R_i and R_j represent customer interaction vectors for products i and j.

Products with higher cosine similarity values are considered behaviorally similar because they are purchased by similar customer groups. These similarity relationships provide additional behavioral information for warehouse storage optimization.

The generated customer purchase relationships are later integrated with Graph Convolutional Network embeddings to form hybrid feature representations. This integration enables the proposed framework to jointly learn structural product relationships and customer purchasing behavior for intelligent warehouse storage assignment.

Overall, customer purchase relationship modeling enables the framework to capture realistic warehouse picking patterns and improves storage optimization by grouping frequently co-picked and behaviorally similar products into nearby warehouse locations.

3.4 Co-Purchase Graph Generation

After modeling customer purchase relationships, a co-purchase graph is generated to represent relational dependencies between products. The co-purchase graph is one of the most important components of the proposed warehouse optimization framework because it converts transactional purchase data into graph-structured data suitable for Graph Neural Network learning.

The graph is represented as:

[G = (V,E)]

where:

- (V) represents product nodes (SKUs)
- (E) represents co-purchase relationships between products

In the generated graph, each SKU is represented as a node, and an edge is created between two products if they appear together in the same customer invoice. The edge weight represents the frequency of co-occurrence between products.

The edge weight between products (i) and (j) is calculated as:

$$w_{ij} = \sum_{k} I(i, j \in Order_k)$$

where:

- w_{ij} represents the co-purchase frequency between products i and j
- $Order_k$ represents customer order k
- $I(i, j \in Order_k)$ returns 1 if both products appear in the same invoice

The graph generation process begins by grouping products using invoice numbers. For each invoice, all possible SKU pairs are generated, and their co-occurrence frequency is counted. The resulting graph is constructed using the NetworkX library and later converted into PyTorch Geometric (PyG) format for Graph Convolutional Network training.

Table 3.1 presents the statistics of the generated co-purchase graph.

Table 3.6: Co-Purchase Graph Statistics

Graph Property	**Value**
Number of SKU Nodes	788
Number of Graph Edges	263,300
Node Feature Dimension	6
Edge Index Shape	[2, 527388]
Edge Weight Count	527,388

The generated graph preserves important structural information regarding product relationships inside the warehouse. Products with strong co-purchase relationships become closely connected in the graph, allowing the Graph Convolutional Network to learn meaningful relational embeddings for warehouse storage optimization.

To visualize the learned storage zones, t-SNE dimensionality reduction is applied to the learned GNN embeddings. The visualization projects high-dimensional SKU embeddings into a two-dimensional space, where products belonging to similar storage zones appear closer together.

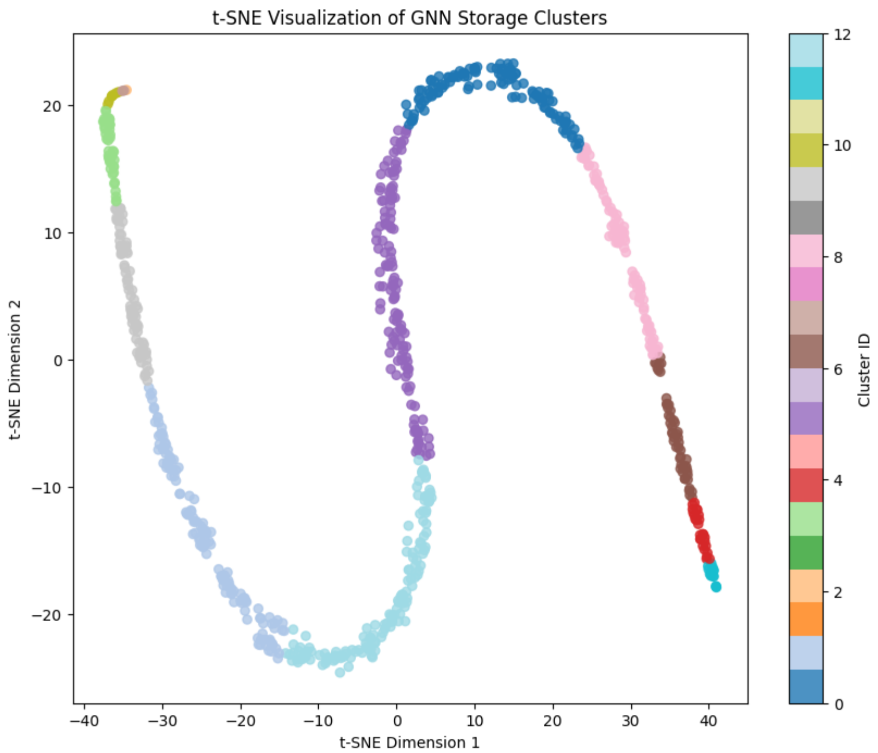

Figure 3.2: t-SNE Visualization of GNN-Based Storage Zone Clusters

The visualization demonstrates clear separation between multiple storage clusters generated by the proposed framework. Products with similar purchasing behavior and strong co-purchase relationships are grouped into nearby embedding regions, indicating that the Graph Neural Network successfully learned meaningful relational representations for warehouse storage optimization.

Overall, the co-purchase graph generation and embedding learning stages transform transactional customer behavior into structured relational representations that enable intelligent warehouse storage assignment using Graph Neural Networks.

3.5 Graph Convolutional Network Based Embedding Learning

After generating the co-purchase graph, Graph Convolutional Network (GCN) based embedding learning is performed to learn meaningful low-dimensional representations of products. The objective of this stage is to capture both SKU demand characteristics and co-purchase relationships through graph-based learning. These learned embeddings are later used for clustering-based warehouse storage optimization.

Unlike traditional machine learning methods that process products independently, Graph Convolutional Networks utilize graph-structured relationships between products. In the proposed framework, products that are frequently purchased together influence each other during graph learning, allowing the model to learn more informative SKU representations.

The generated co-purchase graph and normalized SKU feature matrix are provided as inputs to the GCN model. During each graph convolution layer, neighboring node information is aggregated to update node embeddings. The graph convolution operation is represented as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right)$$

where:

- $\tilde{A}$ represents the adjacency matrix with self-loops
- $\tilde{D}$ represents the degree matrix
- $H^{(l)}$ represents node embeddings at layer l
- $W^{(l)}$ represents trainable weights
- σ represents the ReLU activation function

Since ground-truth storage labels are unavailable, pseudo-labels are generated using K-Means clustering. The GCN model is then trained using these labels with cross-entropy loss and the Adam optimizer.

During training, the loss value gradually decreases, indicating successful learning of product relationships and embedding representations. Figure 3.3 shows the training loss curve of the GCN model.

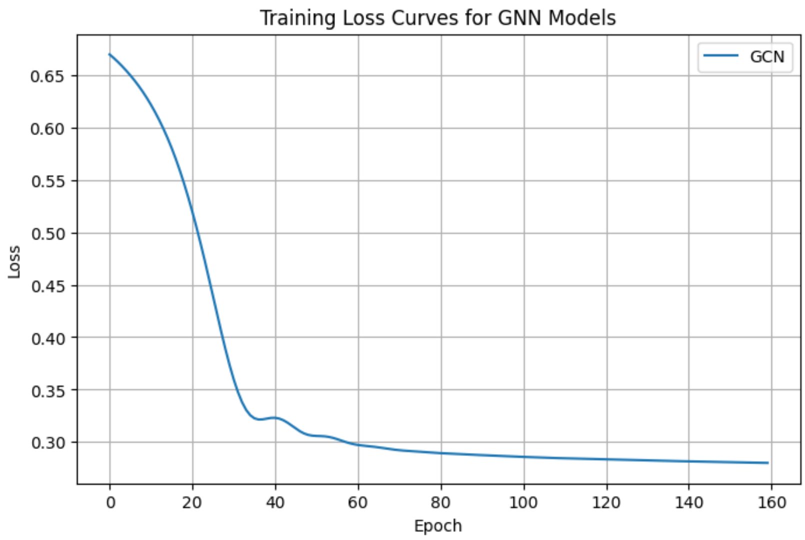

Figure 3.3: Training Loss Curve of the GCN Model

After training, the learned embeddings are evaluated using the Silhouette Score to measure clustering quality. Table 3.7 presents the comparison of different embedding approaches.

Table 3.7: Silhouette Score Comparison

Model	**Silhouette Score**
GCN	0.4202
Hybrid	0.3631
AHC	0.2674

The higher silhouette score achieved by the GCN model indicates better cluster separation and more meaningful embedding learning compared to traditional clustering approaches.

In summary, the GCN-based embedding learning stage allows the framework to learn complex relations among products from customer purchase data and generate meaningful product representations. These learned embeddings provide useful information for clustering related products, and enable more efficient optimization of warehouse storage.

3.6 Collaborative Filtering Similarity Computation

In addition to learning Graph Convolutional Network (GCN), the proposed framework also leverages Collaborative Filtering (CF) to learn product similarity based on customers' purchasing behavior. GCN learns the relations from the co-purchase graph of products , and CF finds products that are often purchased by the same groups of customers . This provides complementary behavioral information that helps to understand better the product associations and supports better warehouse storage optimization.

The collaborative filtering process begins by constructing a User-Item Interaction Matrix from the cleaned transaction dataset. In this matrix, rows represent customers, columns represent products (SKUs), and matrix values represent purchased quantities.

The generated User-Item Interaction Matrix is represented as:

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix}$$

where:

- m represents the number of customers
- n represents the number of SKUs
- r_{ij} represents purchase quantity of SKU j by customer i

Cosine similarity is then applied to measure similarity between products based on customer purchasing behavior:

$$Similarity(i, j) = \frac{R_i \cdot R_j}{||R_i||||R_j||}$$

where R_i and R_j represent customer interaction vectors for products i and j.

Products with higher similarity values are considered behaviorally similar because they are purchased by similar customer groups. These similarity relationships help improve warehouse storage assignment by grouping related products into nearby storage locations.

Table 3.5 presents the statistics of the generated User-Item Interaction Matrix.

Table 3.8: User-Item Interaction Matrix Statistics

Property	Value
Number of Customers	4,273
Number of Selected SKUs	788
Matrix Dimension	4273 ×788
Interaction Type	Purchased Quantity

The generated collaborative filtering similarity features are later combined with Graph Convolutional Network embeddings to form hybrid feature representations for clustering-based warehouse storage optimization.

Overall, Collaborative Filtering enhances the proposed framework by incorporating customer behavioral similarity into the warehouse storage assignment process.

3.7 Hybrid Feature Fusion Strategy

After generating Graph Convolutional Network (GCN) embeddings and Collaborative Filtering (CF) similarity features, the proposed framework combines both representations to create a unified hybrid feature representation for warehouse storage optimization. The objective of hybrid feature fusion is to integrate structural graph relationships and customer purchasing behavior into a single feature space that improves clustering quality and warehouse storage assignment.

The GCN model learns relational product information from the co-purchase graph, while Collaborative Filtering captures behavioral similarity between products based on customer purchasing patterns. Combining both representations enables the framework to utilize complementary information for more effective warehouse optimization.

The output of the GCN learning process is a set of low-dimensional product embeddings, represented as:

$$Z \in \mathbb{R}^{N \times d}$$

where:

- N represents the number of SKUs
- d represents embedding dimension

Similarly, the features obtained from Collaborative Filtering are represented as:

$$CF \in \mathbb{R}^{N \times s}$$

where s denotes the number of similarity-based features generated through Collaborative Filtering. These features capture behavioral relationships between products based on customer purchasing patterns.

To take advantage of both graph-based and behavioral information, the GCN embeddings and Collaborative Filtering features are combined through feature concatenation:

$$X_{Hybrid} = [Z|CF]$$

where:

- Z represents GCN embeddings
- CF represents collaborative filtering similarity features
- X_{Hybrid} represents the final hybrid feature matrix

Before feature fusion, the GCN embedding and Collaborative Filtering similarity features are normalized to bring both features to a similar scale to improve the clustering performance. The obtained hybrid feature representation captures product relationships from various perspectives including co-purchase patterns, neighborhood information from the graph, and customer purchasing behavior.

This hybrid feature matrix is then supplied as input to the K-Means clustering algorithm, to generate the warehouse storage zones. Similar hybrid products are stored in the same zone, so that related products are stored closer together, reducing the picker travel distance when fulfilling orders.

Experimental results obtained in this project demonstrate that the Hybrid framework achieves better clustering quality and lower picker travel distance compared to standalone GCN-based approaches and traditional clustering methods. The Hybrid model achieved an average improvement of 11.53% over Agglomerative Hierarchical Clustering (AHC), outperforming the standalone GNN approach.

Overall, the hybrid feature fusion strategy improves warehouse storage optimization by jointly utilizing graph-based relational learning and customer purchasing similarity information.

3.8 Storage Zone Formation using Clustering

After generating the hybrid feature representation, the next stage of the proposed framework involves forming optimized warehouse storage zones using clustering techniques.

The objective of storage zone formation is to group similar products into nearby warehouse regions so that frequently co-picked products can be picked with minimum travel distance during order fulfillment.

The proposed framework utilizes K-Means clustering on the hybrid feature matrix generated from Graph Convolutional Network (GCN) embeddings and Collaborative Filtering (CF) similarity features. The hybrid representation captures co-purchase relationships, neighboring graph information, SKU demand characteristics, and customer purchasing behavior.

The hybrid feature matrix is represented as:

$$X_{Hybrid} = [Z|CF]$$

where:

- Z represents GCN embeddings
- CF represents collaborative filtering similarity features

The K-Means clustering algorithm partitions products into K storage zones by minimizing intra-cluster distance and maximizing inter-cluster separation. In this project, the products are grouped into 13 warehouse storage clusters.

The clustering objective function is represented as:

$$J = \sum_{i=1}^{K} \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

where:

- K represents the number of clusters
- C_i represents cluster i
- x_j represents hybrid feature vectors
- μ_i represents centroid of cluster i

Products belonging to the same cluster are assigned to nearby warehouse locations. This grouping strategy reduces aisle switching, picker movement, and overall warehouse travel distance during order fulfillment.

The clustering quality is evaluated using the Silhouette Score. Table 3.9 presents the comparison of different clustering approaches.

Table 3.9: Clustering Performance Comparison

Method	Silhouette Score
GCN	0.4202
Hybrid	0.3631
AHC	0.2674

The results show that the proposed GCN and Hybrid approaches achieve better cluster quality compared to traditional Agglomerative Hierarchical Clustering (AHC), leading to more effective warehouse storage zoning.

Overall, clustering-based storage zone formation transforms learned product relationships into practical warehouse storage assignments that improve warehouse operational efficiency and reduce picker travel distance.

3.9 Warehouse Storage Position Assignment

After generating optimized storage zones using clustering, the next stage of the proposed framework involves assigning products to physical warehouse storage locations. The objective of warehouse storage position assignment is to place products belonging to the same storage cluster in nearby warehouse positions so that picker travel distance can be minimized during order fulfillment.

The clustering stage groups products according to their hybrid feature similarity, which includes Graph Convolutional Network (GCN) embeddings and Collaborative Filtering (CF) similarity features. However, clustering alone does not define the exact physical location of products inside the warehouse. Therefore, a storage position assignment strategy is required to map clustered products into warehouse coordinates and storage aisles.

The warehouse setting under consideration in the experiment is depicted as an organized arrangement of different aisles and rack locations along with their respective storage slots. The products allocated the same storage area are placed near one another within the warehouse, facilitating the proximity of similar goods and minimizing picker movements along the aisles while fulfilling orders.

The physical warehouse location assigned to each SKU is depicted as:

$$L_i = (A_i, R_i, S_i)$$

where:

- A_i represents aisle number
- R_i represents rack position

- S_i represents storage slot

Position assignment in the storage system entails the clustering of items into groups, and then assigning positions close to each other in the warehouse to items that belong to the same group. Items that are high in demand and picked together are assigned positions near the entrance and exit areas of the warehouse.

These assignments of SKUs to locations are then used in simulating warehouse routes for different picking scenarios.

Experimental results demonstrate that the Hybrid storage assignment strategy produces more compact warehouse layouts by placing behaviorally similar and frequently co-picked products in nearby storage regions. This reduces aisle switching, picker movement, and overall travel distance during warehouse operations.

Overall, the warehouse storage position assignment stage converts learned product relationships into practical warehouse storage arrangements for efficient warehouse order picking operations.

3.10 Picking Simulation and Routing Evaluation

After assigning products to optimized warehouse storage positions, the effectiveness of the proposed framework is evaluated using warehouse picking simulation and routing analysis. The objective of this stage is to measure how efficiently the generated storage assignments reduce picker travel distance during order fulfillment operations.

The warehouse simulation environment consists of multiple aisles, storage slots, depot locations, SKU coordinates, and customer orders generated from the processed transaction dataset. During simulation, customer orders are replayed, and pickers travel through warehouse aisles to collect required products.

The proposed framework evaluates warehouse performance under three picking strategies:

- Discrete Picking
- Batch Picking
- Zone Picking

Additionally, two standard warehouse routing methods are implemented:

- S-Shape Routing
- Midpoint Routing

The total picker travel distance is calculated using warehouse coordinate positions assigned to products. The total picking distance for a customer order is represented as:

$$D = \sum_{i=1}^{n-1} dist(L_i, L_{i+1})$$

where:

- D represents total picker travel distance
- L_i represents warehouse location of SKU i
- $dist(L_i, L_{i+1})$ represents movement distance between consecutive locations

The proposed framework compares picker travel distance across three different storage optimization approaches:

- Agglomerative Hierarchical Clustering (AHC)
- GNN-based storage optimization
- Hybrid (GCN + CF) storage optimization

Table 3.10 presents the average picker travel distance under different routing and picking strategies.

Table 3.10: Average Picker Travel Distance Comparison

Scenario	**GNN**	**Hybrid**	**AHC**
S-Discrete	161.73	160.24	180.91
S-Batch	367.03	366.90	389.49
S-Zone	165.36	161.03	188.09
M-Discrete	114.56	112.41	125.63
M-Batch	321.05	319.08	345.89
M-Zone	101.73	98.52	122.09

Figure 3.4 shows the comparison of picker travel distance across different warehouse picking scenarios.

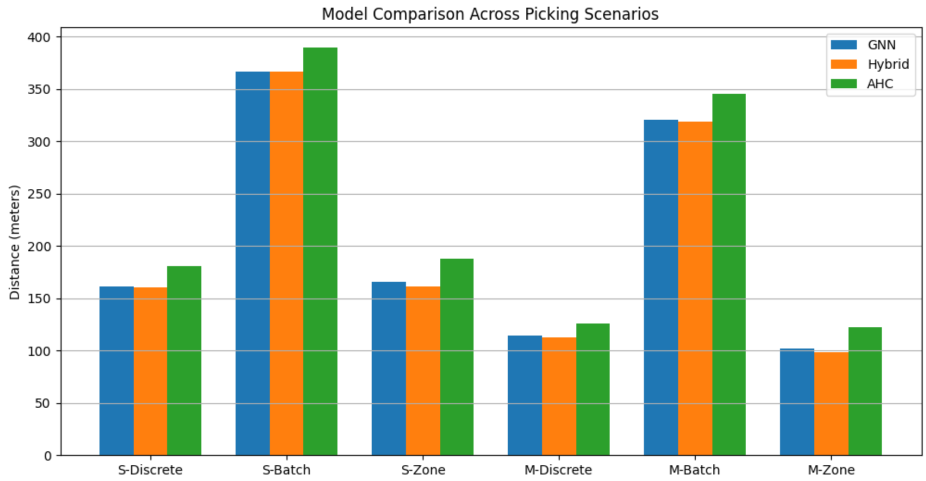

Figure 3.4: Model Comparison Across Picking Scenarios

The results demonstrate that the proposed Hybrid (GCN + CF) framework consistently achieves lower picker travel distance compared to GNN-based and traditional AHC approaches. The reduction in picker movement is achieved by placing frequently co-picked and behaviorally similar products in nearby warehouse locations.

Overall, the simulation results validate the effectiveness of integrating graph-based relational learning and collaborative filtering similarity analysis for intelligent warehouse storage optimization.

3.11 Performance Comparison and Result Analysis

The performance of the proposed warehouse storage optimization framework is evaluated by comparing picker travel distance under different storage assignment approaches, picking strategies, and routing methods. The objective of the evaluation is to analyze how effectively the proposed Hybrid Graph Convolutional Network and Collaborative Filtering (GCN + CF) framework reduces warehouse picker movement compared to traditional clustering approaches.

The proposed framework is compared with the following methods:

- Agglomerative Hierarchical Clustering (AHC)
- GNN-based clustering
- Hybrid (GCN + CF) clustering

The evaluation is conducted using warehouse routing simulation under different picking strategies including Discrete Picking, Batch Picking, and Zone Picking using both S-Shape and Midpoint routing methods.

Table 3.11 presents the average picker travel distance obtained from the simulation experiments.

Table 3.11: Performance Comparison of Storage Optimization Approaches

Scenario	GNN	Hybrid	AHC
S-Discrete	161.73	160.24	180.91
S-Batch	367.03	366.90	389.49
S-Zone	165.36	161.03	188.09
M-Discrete	114.56	112.41	125.63
M-Batch	321.05	319.08	345.89
M-Zone	101.73	98.52	122.09

Figure 3.5 shows the comparison of picker travel distance across different warehouse picking scenarios.

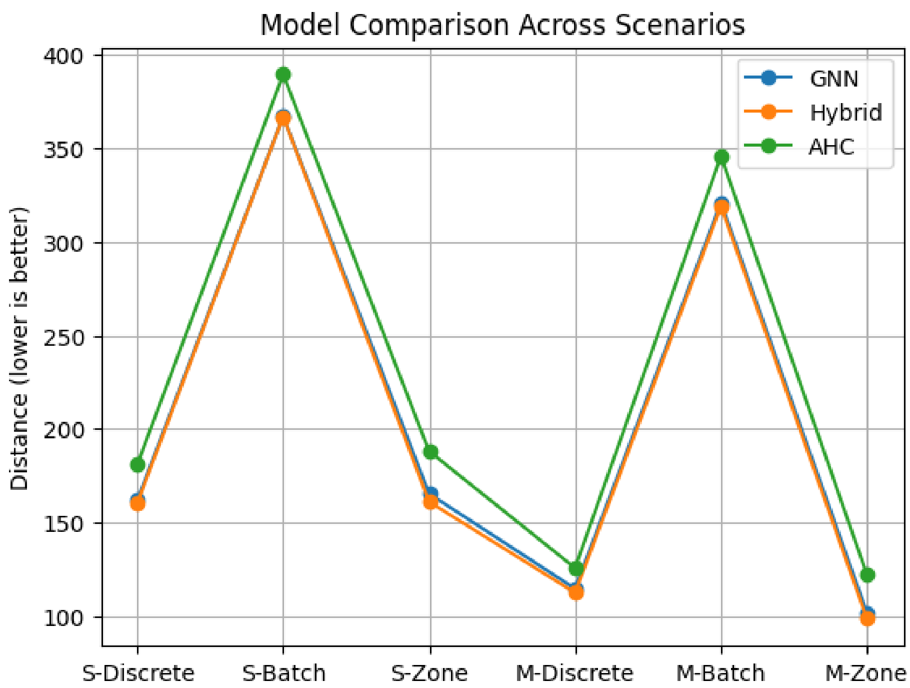

Figure 3.5: Comparison of Picker Travel Distance Across Different Picking Scenarios

The experimental results show that the proposed Hybrid (GCN + CF) framework consistently achieves lower picker travel distance compared to GNN-based and traditional AHC approaches. The Hybrid framework performs better because it combines graph-based co-purchase relationships with customer purchasing similarity information.

The traditional AHC method produced the highest travel distance across all scenarios because it does not effectively capture relational product behavior. The standalone GNN approach improved warehouse routing performance by learning structural relationships from the co-purchase graph. However, the Hybrid framework achieved the best overall performance by integrating both graph learning and collaborative filtering similarity features.

The percentage improvement of the Hybrid framework over AHC is calculated as:

$$Improvement = \frac{D_{AHC} - D_{Hybrid}}{D_{AHC}} \times 100$$

Experimental analysis shows that the proposed Hybrid framework achieved approximately 11% average reduction in picker travel distance compared to traditional clustering approaches.

Overall, the obtained results validate the effectiveness of integrating Graph Neural Networks and Collaborative Filtering for intelligent warehouse storage optimization and efficient warehouse routing.

3.12 Discussion and Interpretation of Results

The experimental results demonstrate that the proposed Hybrid Graph Convolutional Network and Collaborative Filtering (GCN + CF) framework is effective in reducing picker travel distance and improving warehouse storage efficiency. The obtained results consistently show that the Hybrid approach outperforms both the standalone GNN model and the traditional Agglomerative Hierarchical Clustering (AHC) method across all evaluated picking scenarios.

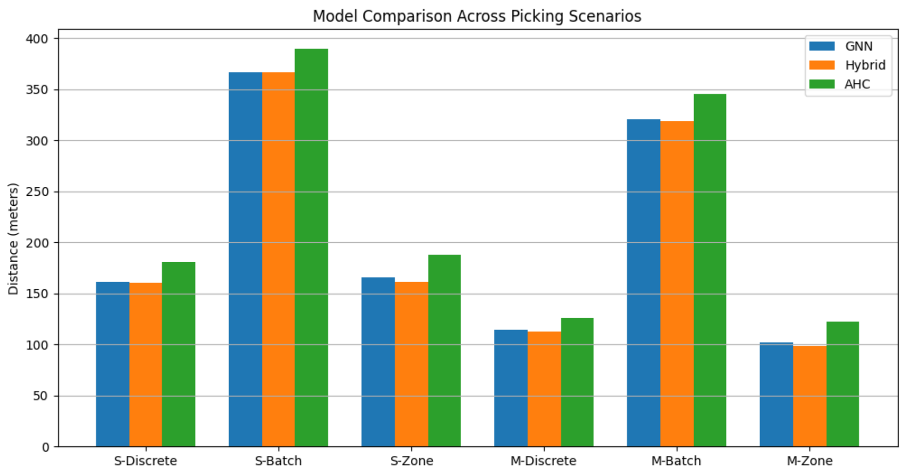

Figure 3.6: Performance Comparison of GNN, Hybrid, and AHC Approaches

As shown in Figure 3.6, the Hybrid framework achieved the lowest picker travel distance in all six simulation scenarios. The best performance was observed in the M-Zone scenario, where the Hybrid model reduced travel distance to 98.52 meters compared to 122.09 meters for AHC.

The Hybrid approach is better because it can combine graph-based relational learning and customer purchasing behavior. The GNN module learns the co-purchase relations among the products and Collaborative Filtering learns the similarities based on the customer preferences. The framework merges the two information sources to generate more meaningful storage zones and better product placement in the warehouse.

The standalone GNN model also outperformed AHC by learning structural relationships from the co-purchase graph. However, the Hybrid framework was able to achieve further improvements by including customer behavior information, leading to more effective clustering, shorter picking routes and reduced picker travel distance.

Another important observation is that Midpoint routing generally produced shorter travel distances than S-Shape routing. This suggests that reducing unnecessary aisle travel can greatly improve warehouse picking efficiency. In addition, Zone Picking scored better because related products were stored in close locations.

The overall results show that the combination of Graph Neural Networks with Collaborative Filtering is effective for warehouse storage optimization. The Hybrid framework proposed was successful in reducing the picker travel distance, improving storage organization, and improving the overall warehouse efficiency by combining product relationships with customer purchasing behavior.

3.13 Limitations and Future Improvements

While the proposed Hybrid GCN and Collaborative Filtering framework achieved significant improvements in reducing picker travel distance, several limitations should be considered when interpreting the results.

First, the framework is developed using historical transaction data and assumes a relatively static warehouse environment. In practice, customer purchasing behavior, seasonal trends, and inventory availability continuously change over time. As a result, storage assignments generated from historical data may become less effective if they are not updated regularly to reflect current demand patterns.

Secondly, it should be noted that the optimization algorithm mainly revolves around reducing the distances travelled by the picker. Although travel distance constitutes an essential element of warehouse efficiency, there are other operational parameters that have been omitted from consideration within the proposed framework. These include such parameters as storage capacity constraints, replenishment requirements, product characteristics, distribution of workload among the pickers, and aisle congestion that might affect warehousing.

One more weakness of the research under consideration concerns the simulated warehouse environment where experiments were conducted. Indeed, the simulation model used in this research is quite simple and cannot reflect some important aspects of real-world operations. In particular, the model does not take into account such aspects as picking performed simultaneously by several workers, congestion issues, use of automatic guided vehicles, and inventory flows.

Moreover, the Graph Convolutional Network (GCN) has been trained using pseudo-labels that have been created using K-Means clustering since there were no real storage labels from the warehouse. Even if this technique allows for semi-supervised learning, the quality of the learned product embeddings relies greatly on the quality of the clustering performed. The mistakes made during the clustering may impact how effective the embeddings will be.

Nevertheless, the framework presented above can be considered as a good starting point for further research in the field. For example, one way to extend the research could be designing an approach for creating storage labels dynamically based on the real time-demand information and inventory data available in the database.

There might also be an option to use more complex models such as GraphSAGE, Graph Attention Networks (GAT), and Temporal Graph Networks (TGN) in order to enhance representation learning, scalability, and adaptability of the graph network.

Moreover, reinforcement learning methods can be incorporated within the framework in order to facilitate dynamic storage allocation and routing decisions in the warehouse. Through reinforcement learning, which involves interactions between the agents and the warehouse environment, even more effective operations and decisions can be made.

Furthermore, the framework can incorporate new warehouse technologies such as AMRs (autonomous mobile robots), robotic picking systems, and intelligent warehouse management systems. This will enhance the applicability of the suggested framework in modern-day warehouses.

Finally, future research may investigate multi-objective optimization methods that simultaneously consider multiple operational goals, including picker travel distance, picking time, warehouse congestion, labor utilization, and overall operational cost. Optimizing these objectives together would provide a more comprehensive and realistic solution for large-scale warehouse management.

Overall, although several challenges remain, the proposed Hybrid GCN and Collaborative Filtering framework demonstrates considerable potential for intelligent warehouse storage optimization. The findings of this study provide a valuable foundation for future developments and support the application of graph-based learning techniques in next-generation warehouse systems.

3.14 Chapter Summary

In this chapter, we introduced the proposed Hybrid Graph Convolutional Network and Collaborative Filtering (GCN + CF) based warehouse storage optimization framework that is aimed to minimize the picker travel distance and enhance the warehouse operational efficiency.

This chapter presents the complete workflow of the proposed warehouse storage optimization framework. The process started with preprocessing transaction data and SKU feature engineering, then analyzed customer purchasing behavior to build a co-purchase graph. We use a Graph Convolutional Network (GCN) to learn product representations from the graph structure and Collaborative Filtering to learn product similarities based on customer purchase patterns.

The learned GCN embeddings and Collaborative Filtering features were concatenated to form a hybrid product representation. The K-Means clustering algorithm then used these hybrid features to create storage zones, and products were allocated to warehouse locations based on the clusters. The idea was that related products would be stored more closely together which would increase storage efficiency.

Validations were done with the help of warehouse picking simulations with varying picking techniques and routing strategies. It was found that the Hybrid (GCN + CF) method successfully produced the least picker travel distance relative to the Agglomerative Hierarchical Clustering (AHC) method and the standalone GNN method. This framework integrates the graph-based relationships among products with customer purchasing behavior and leads to efficient storage allocations and picking operations.

Moreover, the chapter discussed the drawbacks of the current model and mentioned potential research areas for further development of the concept, which includes dynamic storage allocation, advanced Graph Neural Network models, reinforcement learning methods, and automation of the warehouses. Finally, based on the findings of the paper, it can be concluded that the proposed framework integrating Graph Neural Networks with Collaborative Filtering is an effective technique.

Chapter 4

Conclusions and Future Works

4.1 Conclusions

This project proposed a Hybrid Graph Convolutional Network and Collaborative Filtering (GCN + CF) framework for warehouse storage optimization, in order to minimize picker travel distance, and improve overall warehouse efficiency. The framework used historical data from retail transactions to understand customer purchasing behavior, patterns in product demand, and relationships between products that are often purchased together.

First, transaction data was cleaned and processed to extract meaningful SKU level features. A co-purchase graph was created, which had products as nodes and co-purchase relationships as weighted connections. A Graph Convolutional Network (GCN) used this graph to learn product embeddings that reflect structural relationships between products. Simultaneously, Collaborative Filtering was used to determine product similarities based on customer buying behavior and user-item interactions.

A hybrid feature fusion strategy is used to fuse the learned graph embeddings and collaborative filtering similarity features. Then, K-Means clustering was used to create optimized storage zones in the warehouse and products were assigned to the nearest warehouse locations according to the clusters generated.

The proposed framework was assessed through warehouse routing simulations with different picking strategies (Discrete Picking, Batch Picking and Zone Picking) and routing techniques (S-Shape, Midpoint). The results showed that the Hybrid (GCN + CF) approach always achieved lower picker travel distances than both Agglomerative Hierarchical Clustering (AHC) and the standalone GNN-based approach for all scenarios tested.

The results indicate that the storage assignments based on graph-based product relationships along with customer purchasing similarity are more effective. The framework helped to reduce picker movement, improve storage organization and better overall warehouse efficiency by placing similar products within close proximity. The results show the potential of combining Graph Neural Networks and Collaborative Filtering to enable smart warehouse storage optimization and more efficient order fulfillment operations.

4.2 Future Works

The proposed Hybrid GCN and Collaborative Filtering framework yielded promising results for warehouse storage optimization. Nevertheless, there are various opportunities to improve its performance and applicability in real-world warehouse settings.

Potential future research may include developing dynamic approaches for optimizing storage space assignments where the assignments would be adjusted based on transaction information obtained at any particular time and based on varying customer demand trends. Customer buying trends may change due to seasonality, economic conditions prevailing in the markets and customer demands, and thus it might be beneficial to introduce a mechanism that could adapt storage assignments and keep optimal performance levels in warehouses.

Further investigation may also cover the use of advanced GNN architecture like GraphSAGE, Graph Attention Networks (GAT) and Temporal Graph Networks (TGN). Such architecture may allow for modeling the relationships between products better and make the approach scalable, allowing one to work with big data. Furthermore, it may allow for modeling the dynamic aspect of customer behavior.

A possible area of research may also cover introducing machine learning agents and training them through reinforcement learning. Such agents may learn how to assign proper storage locations and how to route products.

Moreover, the existing model can be modified to account for the actual constraints faced by warehouses in day-to-day operations. They may be the limitations on storage capacity, replenishment requirements, size of products, workload distribution amongst pickers and warehouse congestion. Considering such constraints will ensure that the optimization process is closer to reality and is more applicable to practice.

In addition, further research may examine the combination of the model under discussion with other systems of the warehouse. For example, these could be AMRs, robotic picking and intelligent warehouse management systems which would lead to the development of more advanced systems of warehouse operations due to widespread automation.

Another possible direction of further study could be examining the multi-objective optimization approaches considering several performance criteria at once. Such criteria could include the distance traveled by pickers, picker's time on task, warehouse congestion, workforce utilization and operational costs.

In summary, these research directions in the future can help in enhancing the scalability, flexibility, and effectiveness of the framework, thus ensuring that it can be applied effectively in larger scale warehouses.

References

[1] L. Zhou, L. Sun, Z. Li, W. Li, N. Cao, and R. Higgs, "Study on a storage location strategy based on clustering and association algorithms: L. zhou et al." *Soft Computing*, vol. 24, no. 8, pp. 5499–5516, 2020.

[2] H. Kalbha, A. Khiat, A. Bahnasse, and H. Ouajj, "Enhancing warehouse efficiency with time series clustering: A hybrid storage location assignment strategy," *IEEE Access*, vol. 12, pp. 1–15, 2024.

[3] A. Zarinchang, K. Lee, I. Avazpour, J. Yang, D. Zhang, and G. K. Knopf, "Adaptive warehouse storage location assignment with considerations to order-picking efficiency and worker safety," *Journal of Industrial and Production Engineering*, vol. 41, no. 1, pp. 40–59, 2024.

[4] N. Boysen, R. de Koster, and F. Weidinger, "Dynamic storage location assignment in warehouses using deep reinforcement learning," *Computers & Industrial Engineering*, vol. 165, p. 107930, 2022.

[5] Y. Liu, N. Boysen, and D. Briskorn, "Supervised learning and tree search for real-time storage allocation in robotic mobile fulfillment systems," *Transportation Science*, vol. 56, no. 4, pp. 1187–1205, 2022.

[6] J. Li, R. Huang, and J. B. Dai, "Joint optimisation of order batching and picker routing in the online retailer's warehouse in china," *International Journal of Production Research*, vol. 55, no. 2, pp. 447–461, 2017.

[7] M. Özbaltan, S. Çaşka, M. Yıldırım, C. Şeker, F. E. Aysal, H. S. Bıçakcı Yeşilkaya, M. Demir, and E. Kuzu, "Path optimization for cluster order picking in warehouse robotics using hybrid symbolic control and bio-inspired metaheuristic approaches," *Biomimetics*, vol. 10, no. 10, p. 657, 2025.

[8] Z. Wu, S. Pan, F. Chen, and G. Long, "A comprehensive survey on graph neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 4–24, 2021.

[9] Y. Tu, W. Li, X. Song, K. Gong, L. Liu, Y. Qin, and M. Liu, "Using graph neural network to conduct supplier recommendation based on large-scale supply chain," *International Journal of Production Research*, vol. 62, no. 24, pp. 8595–8608, 2024.

[10] C. Cvancara, "Graph neural network based learning for facility location optimization," *Preprints*, 2024.